

Project		
ORCA_TOP Physical Design		
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Abstract

This report presents the end-to-end physical design implementation of the ORCA_TOP module, a 32-bit RISC-V processor core, using Synopsys Fusion Compiler in the SAED 32nm technology node. The design consists of approximately 53,000 standard cells and 40 hard macros.

An R-shaped non-rectangular floorplan was adopted with 75% core utilization to balance area efficiency and routing feasibility. A hierarchical power mesh implemented across M6–M9 ensured reliable power distribution within dual-voltage domains. Clock Tree Synthesis was performed with a 100 ps skew target, and post-route SI-aware optimization achieved approximately 0.9 ns of setup slack recovery.

Post-route analysis indicates stable congestion and density distribution under high utilization conditions. While minor DRC and LVS issues remain for complete signoff closure, the design demonstrates successful timing convergence and practical physical design optimization within a modern ASIC backend flow.

1. Introduction

The transformation of a gate-level netlist into a physical silicon layout is a multi-objective optimization process involving Power, Performance, and Area (PPA). As technology scales into deep-submicron nodes, physical implementation must address routing congestion, signal integrity degradation, clock skew control, and multi-voltage domain integration.

This project focuses on the end-to-end physical realization of the ORCA_TOP core, a 32-bit RISC-V processor implemented in the SAED 32nm technology node using Synopsys Fusion Compiler. The design integrates approximately 53,000 standard cells, 40 hard macros, and two distinct voltage domains (VDD and VDDH), requiring coordinated floorplanning and power distribution strategies.

A structured backend design flow was executed, including:

- Floorplanning & I/O Assignment to define core boundaries and routing channels.
- Power Delivery Network (PDN) construction for stable multi-voltage operation.
- Timing-driven Placement for optimized cell distribution and congestion control.
- Clock Tree Synthesis (CTS) for skew and transition management.
- Detailed Routing & Post-Route Optimization to refine timing and signal integrity.

The primary objective of this implementation is to approach timing convergence within a 16.7 ns clock period while maintaining a 100 ps clock skew target and structural correctness under 75% core utilization, demonstrating practical optimization strategies in a realistic ASIC backend environment.

2. Design Specification

The ORCA_TOP module is a complex hierarchical design based on the RISC-V architecture. The design parameters are summarized below:

- **Technology Node:** saed32nm (32/28nm)
- **Cell Count:** ~53,000 Standard Cells
- **Macro Integration:** 40 Hard Macros (SRAMs, Register Files)
- **Interconnect Stack:** 9 Metal Layers (M1 - M9)
- **Power Domains:** 2 Voltage Domains (VDD and VDDH)
- **I/O Capacity:** 240 Ports

3. Physical Implementation Flow

3.1 Floorplanning and I/O Assignment

3.1.1 Initialize Core Area

The core initialization phase adopted an **R-shaped floorplan** rather than a conventional rectangular boundary. This configuration was selected to improve macro placement efficiency, particularly by utilizing concave regions to reduce global interconnect length toward the central logic area.

A target core utilization of **0.75** was defined to balance placement density and routing resources, providing sufficient whitespace for congestion mitigation and post-placement optimization.

Additionally, a **10 μm core offset** was applied between the core boundary and the IO boundary to ensure adequate routing channels near the periphery and to support robust power ring implementation.

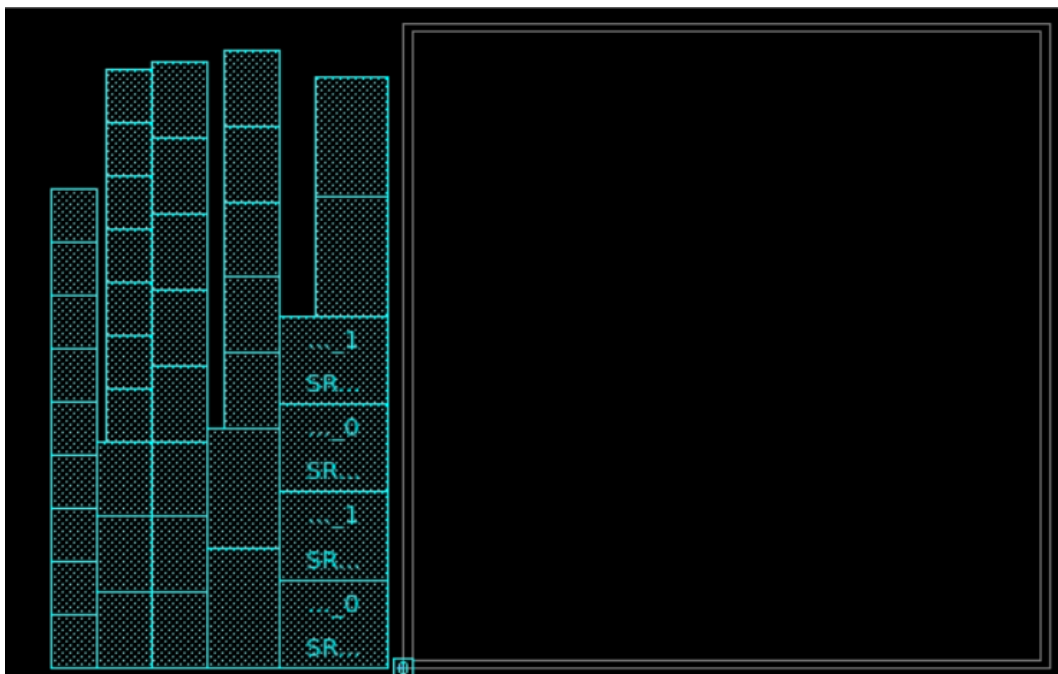


Figure 1: Initialize floorplan

3.1.2 I/O Pin Placement

During the I/O pin placement stage, previous pin constraints were cleared before defining new assignments. Input ports were placed on **M5/M7 (Side 1)**, while output ports were assigned to **M5/M7 (Side 3)** to achieve directional separation and reduce boundary congestion.

Pin offsets were controlled to ensure balanced distribution along the die boundary. Automatic pin placement was then executed to optimize alignment and routing accessibility.

Separating input and output ports across opposite sides and utilizing upper metal layers improves routability and mitigates edge congestion.

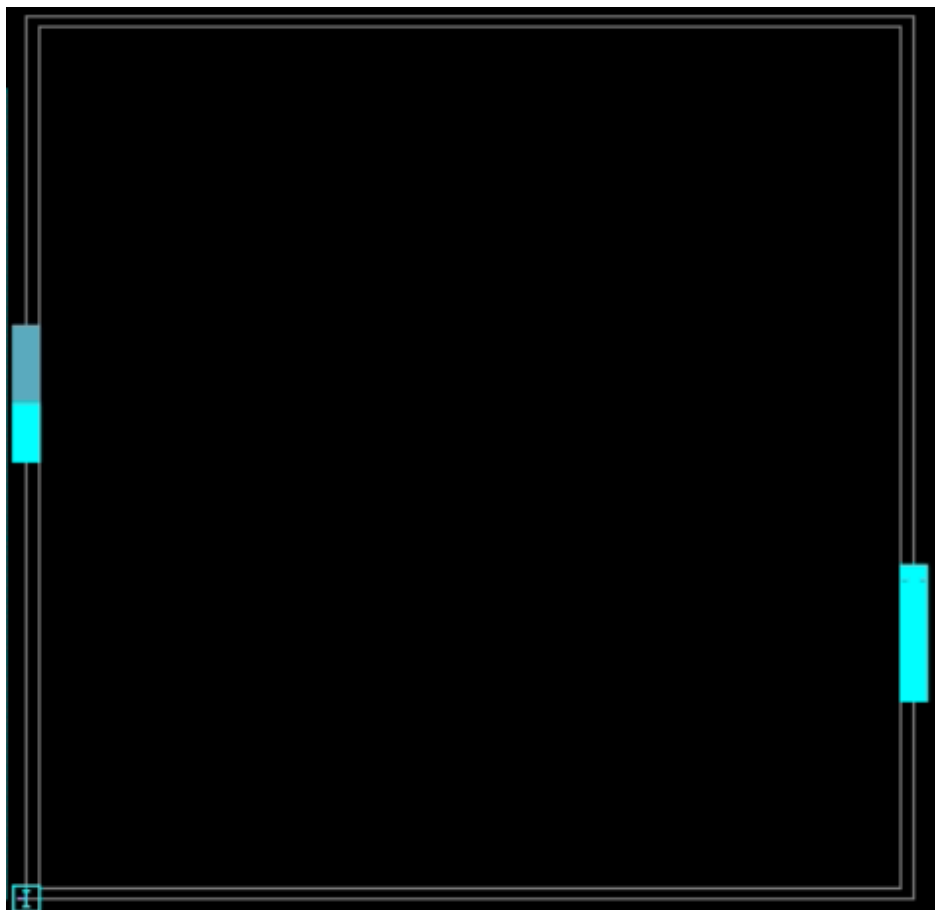


Figure 2: I/O Port placement

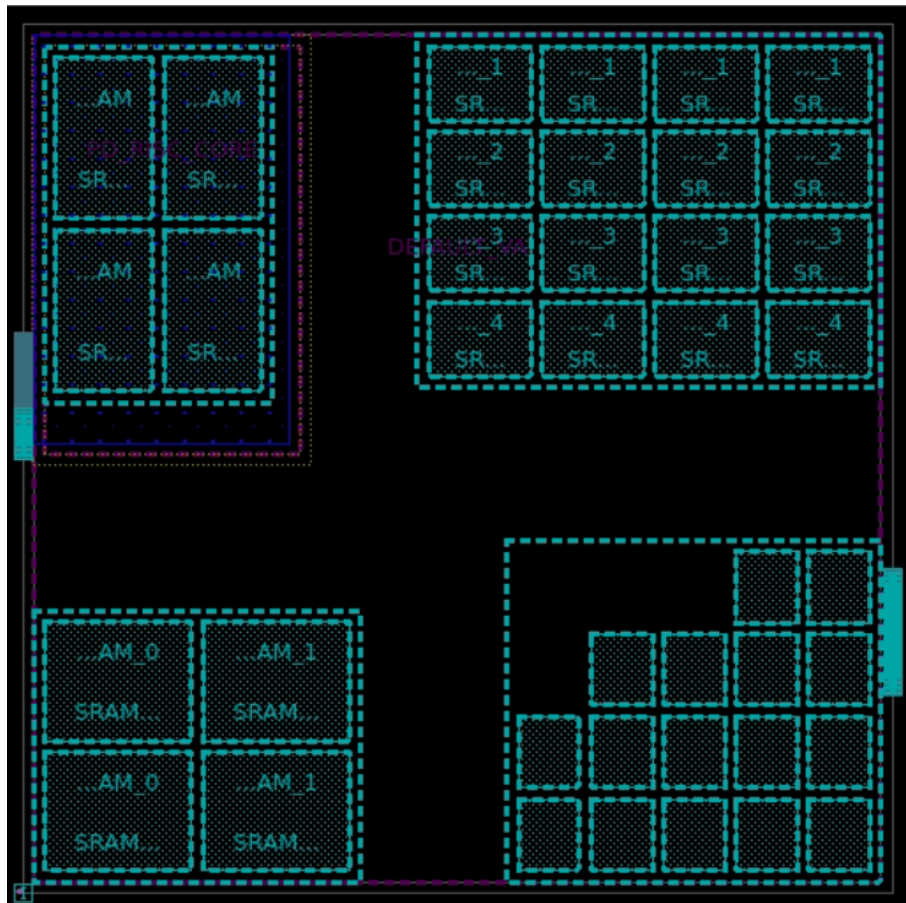


Figure 3: Macro placement & Voltage Area

3.1.3 Sanity Check (Floorplaning)

- **Legality:** Verified with `check_legality` to ensure zero overlaps or site-row violations.

```
fc_shell> check_legality -verbose
Information: The command 'check_legality' cleared the undo history. (UNDO-016)

*****

running check_legality

*****

Report : Legality
*****

VIOLATIONS BY CATEGORY:
  MOVABLE  APP-FIXED  USER-FIXED  DESCRIPTION
      0           0           0  Two objects overlap.
      0           0           0  A cell violates a pnet.
      0           0           0  A cell is illegal at a site.
      0           0           0  A cell is not aligned with a site.
      0           0           0  A cell has an illegal orientation.
      0           0           0  A cell spacing rule is violated.
      0           0           0  A layer rule is violated.
      0           0           0  A cell is in the wrong region.
      0           0           0  Two cells violate cts margins.
      0           0           0  Two cells violate coloring.

      0           0           0  TOTAL

TOTAL 0 Violations.
```

Figure 4: Legality Report

- **Core Utilization Report :**

```
fc_shell> report_utilization
*****
Report : report_utilization
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Sat Feb 14 08:44:53 2026
*****
Utilization Ratio:                0.7068
Utilization options:
- Area calculation based on:      site_row of block ORCA_TOP
- Categories of objects excluded: hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                        586645.6781
Total Capacity Area:              247983.8036
Total Area of cells:              175285.9124
Area of excluded objects:
- hard_macros      :              270039.1823
- macro_keepouts   :              68622.6922
- soft_macros      :              0.0000
- io_cells         :              0.0000
- hard_blockages   :              0.0000
Total Area of excluded objects:    338661.8745
Ratio of excluded objects:         0.5773

Utilization of site-rows with:
- Site 'unit':                0.7068

0.7068
fc_shell>
```

Figure 5: Utilization Report

- **Checking DRC : No errors found**

```
fc_shell> check_pg_drc -ignore_std_cells -do_not_check_shapes_in_hier_blocks
Command check_pg_drc started at Mon Feb 23 23:47:50 2026
Command check_pg_drc finished at Mon Feb 23 23:47:52 2026
CPU usage for check_pg_drc: 1.84 seconds ( 0.00 hours)
Elapsed time for check_pg_drc: 2.01 seconds ( 0.00 hours)
No errors found.
fc_shell>
```

Figure 6: Checking DRC (Floorplaning)

3.2 Power Delivery Network (PDN)

3.2.1 Power Planning

To ensure electrical integrity across the 53k-cell fabric, a **Hierarchical Power Mesh** was constructed.

- **Global Mesh:** Constructed on top metal layers (M6-M9) to minimize resistance.
- **Macro Power Rings:** Dedicated rings on M5/M6 were implemented around each of the 40 hard macros to prevent localized voltage droop.

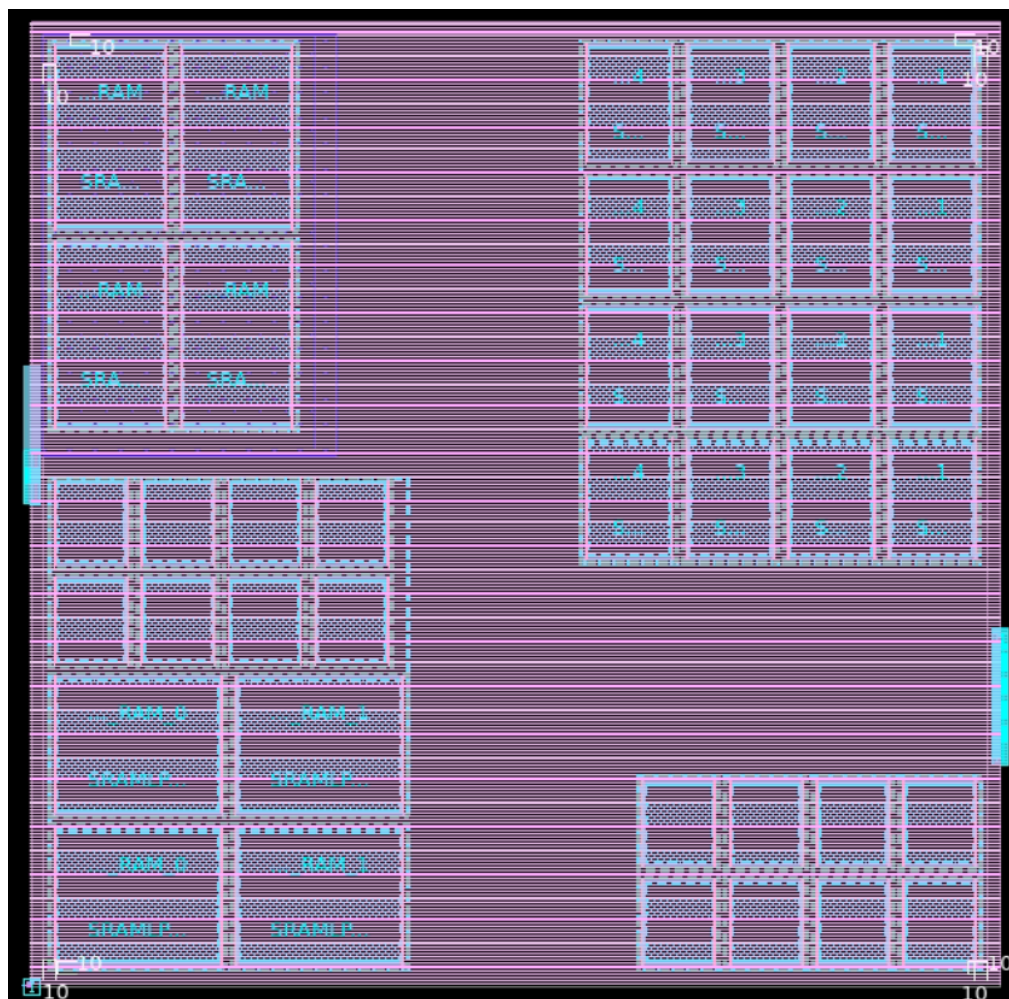


Figure 7: Power routing

3.2.2 Sanity check (Power Routing)

- **Verification:** The `check_pg_connectivity` command confirmed **100% connectivity** with zero floating shapes.

```
fc_shell> check_pg_connectivity
Checking secondary net through power switch is enabled.
Secondary net will be checked together from primary net. They will be treated as the same net
Primary Net : VDD      Secondary Net:
Primary Net : VDDH     Secondary Net:
Primary Net : VSS      Secondary Net:
Loading cell instances...
Number of Standard Cells: 80257
Number of Macro Cells: 40
Number of IO Pad Cells: 0
Number of Blocks: 0
Loading P/G wires and vias...
Number of VDD Wires: 3626
Number of VDD Vias: 67372
Number of VDD Terminals: 0
*****Verify net VDD connectivity*****
Number of floating wires: 721
Number of floating vias: 0
Number of floating std cells: 52877
Number of floating hard macros: 0
Number of floating I/O pads: 0
Number of floating terminals: 0
Number of floating hierarchical blocks: 0
*****
Loading cell instances...
Loading P/G wires and vias...
Number of VDDH Wires: 504
Number of VDDH Vias: 1808
Number of VDDH Terminals: 0
```

Figure 8: Connectivity Results

- **Checking pin placement :** No violations found.

```
fc_shell> check_pin_placement -wire_track true
Warning: Didn't find any enabled planning block for check_pin_placement. (DP)
0 total number of pre-routes loaded
0 total number of vias loaded
0 total number of edges loaded
----- Start Of Missing Pin Check -----
----- End Of Missing Pin Check -----

----- Start Of Wire Track Check -----
----- End Of Wire Track Check -----

----- Start Of Technology Spacing Check -----
----- End Of Technology Spacing Check -----

----- Start Of Pin Short Check -----
----- End Of Pin Short Check -----

No violation has been found
***Summary***
-----
Type of Violation | Count
-----+-----
Missing Pins      | 0
Pins Off Track    | 0
Pin Short         | 0
Technology Spacing | 0
-----+-----
Total Violations  | 0
-----
```

Figure 9: Checking pin placement

- **Checking power/ground missing vias** : No missing vias found.

```
fc_shell> check_pg_missing_vias
Check net VDD vias...
Number of missing vias: 0
Checking net VDD vias took 0 seconds.
Check net VDDH vias...
Number of missing vias: 0
Checking net VDDH vias took 0 seconds.
Check net VSS vias...
Number of missing vias: 0
Checking net VSS vias took 1 seconds.
Overall runtime: 1 seconds.
fc_shell>
```

Figure 10: Checking power/ground missing vias

- **Checking legality**: No violations found.

===== 2D Rule Check Legality Report =====				
Legality Rules	Violating Cells			Total
	Moveable	AppFixed	UserFixed	
overlap	0	0	0	0
spacing_rule	0	0	0	0
cell_on_site	0	0	0	0
legal_orient	0	0	0	0
va_bound	0	0	0	0
pg_drc	0	0	0	0
Total Violations	0	0	0	0
=====				

Figure 11: Checking legality

3.3 Placement

3.3.1 Placement Optimization Strategy

Routing layers were constrained from **M1 to M8** to control resource usage during placement. Tie cell insertion was enabled with a maximum fanout of 8 to ensure safe constant connections.

Routing-Driven Estimation (RDE), along with timing-aware and congestion-aware placement, was activated to balance performance and routability. Scan DEF was loaded before placement, and scan chain reordering was enabled.

High-effort optimization and advanced legalization were applied. Clock-aware and power-aware placement were enabled, while CCD optimization and clock gate latency estimation were disabled. The `place_opt` flow was then executed, followed by automatic power and ground reconnection.

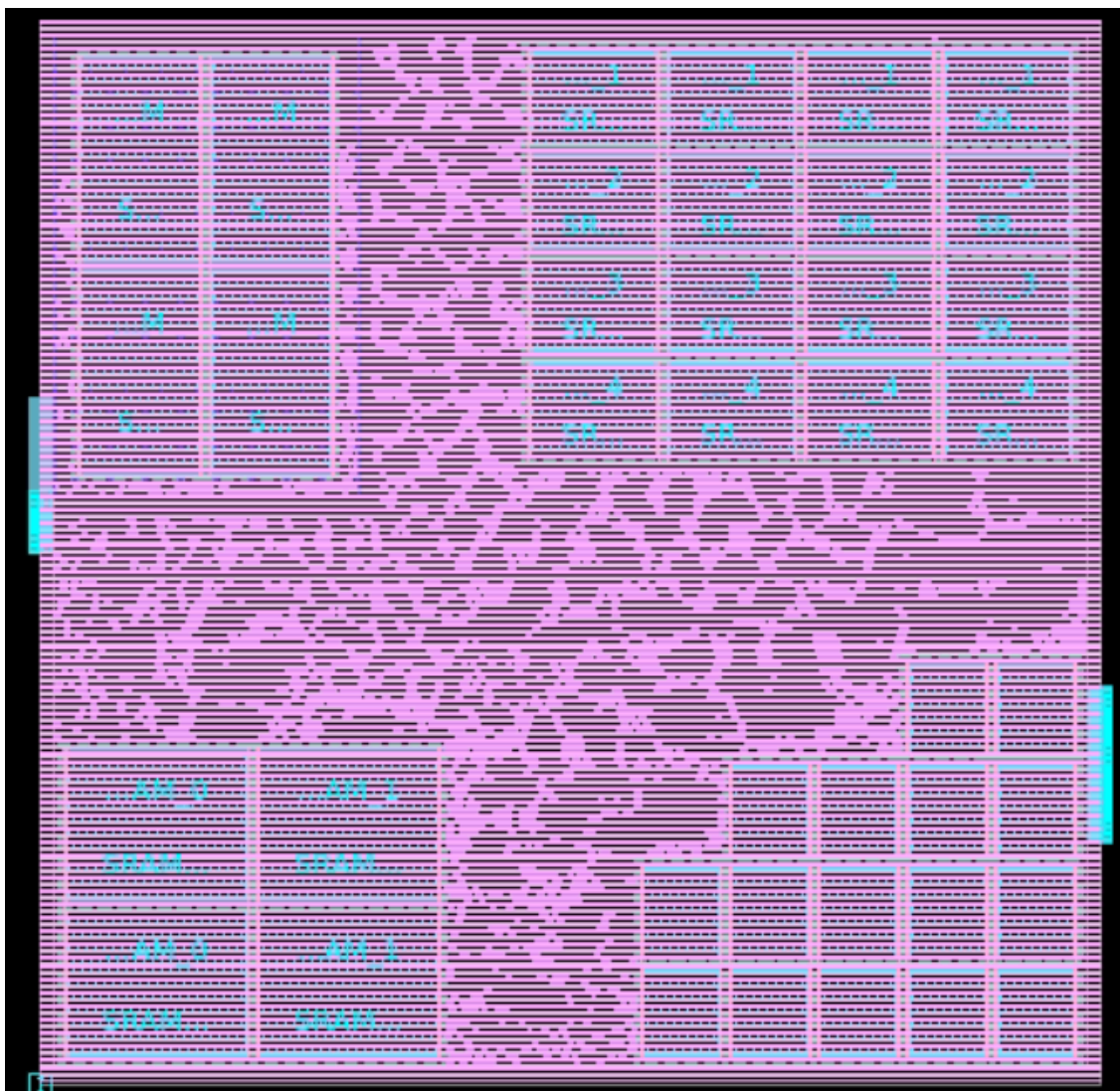


Figure 12: Placement views

3.3.2 Placement Quality Analysis

- **Cell Density:** Almost green (good), a little bit yellow (warning).

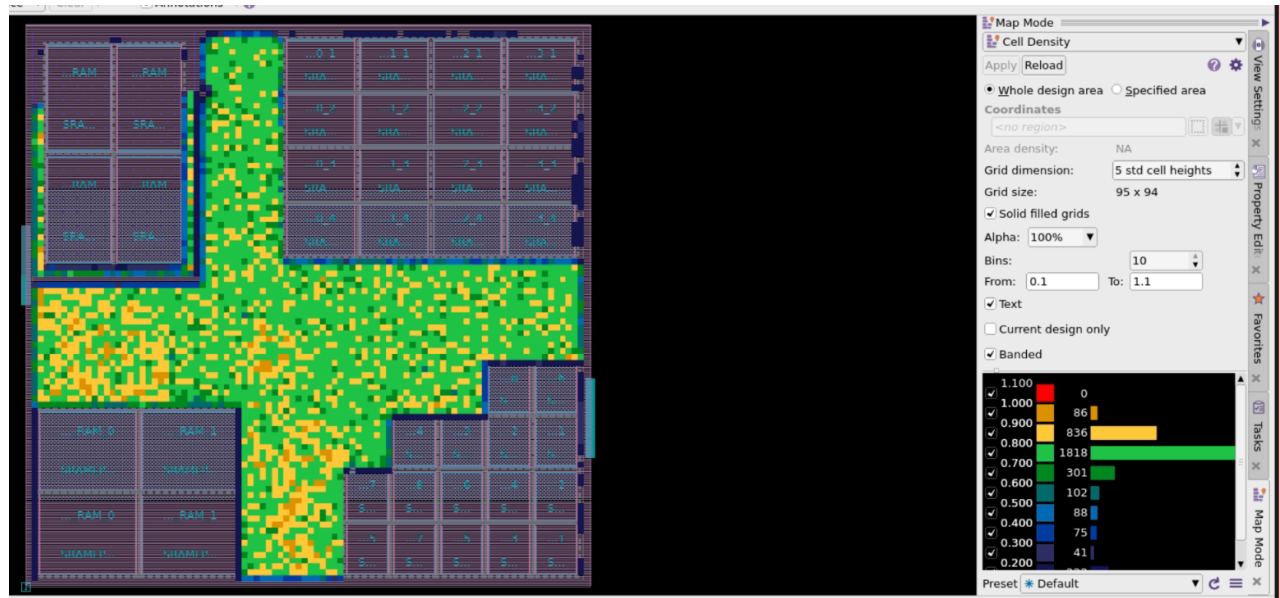


Figure 13: Cell Density

- **Pin Density:** All is blue and green (Very good & good).

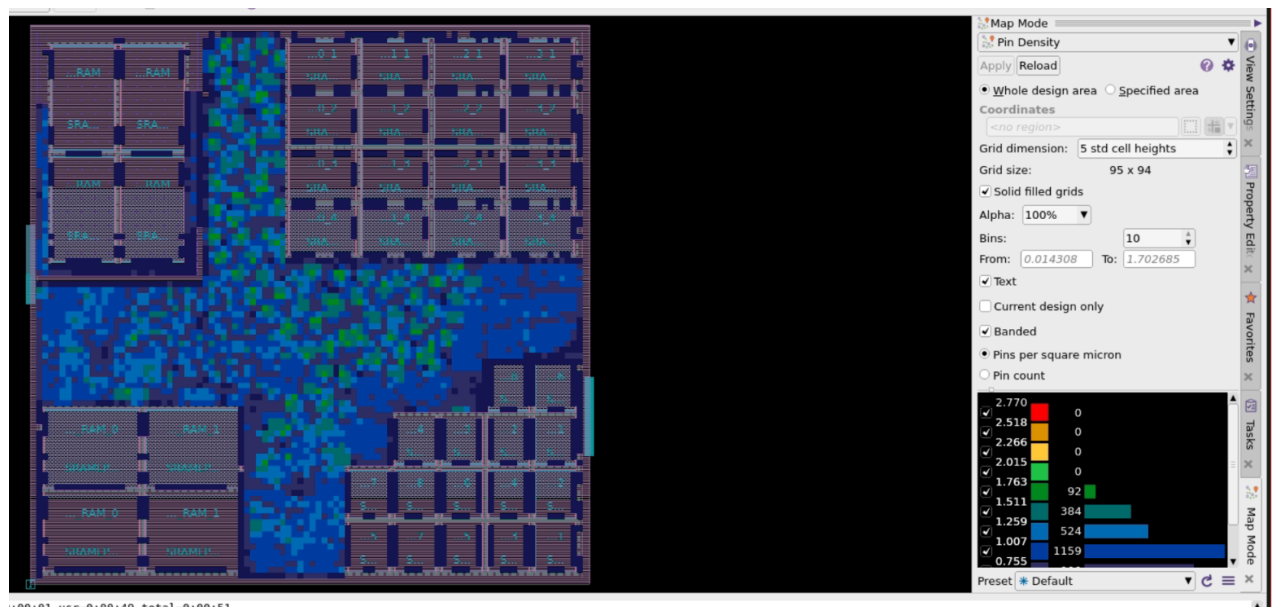


Figure 14: Pin Density

- **Global Route Congestion: No congestions found.**

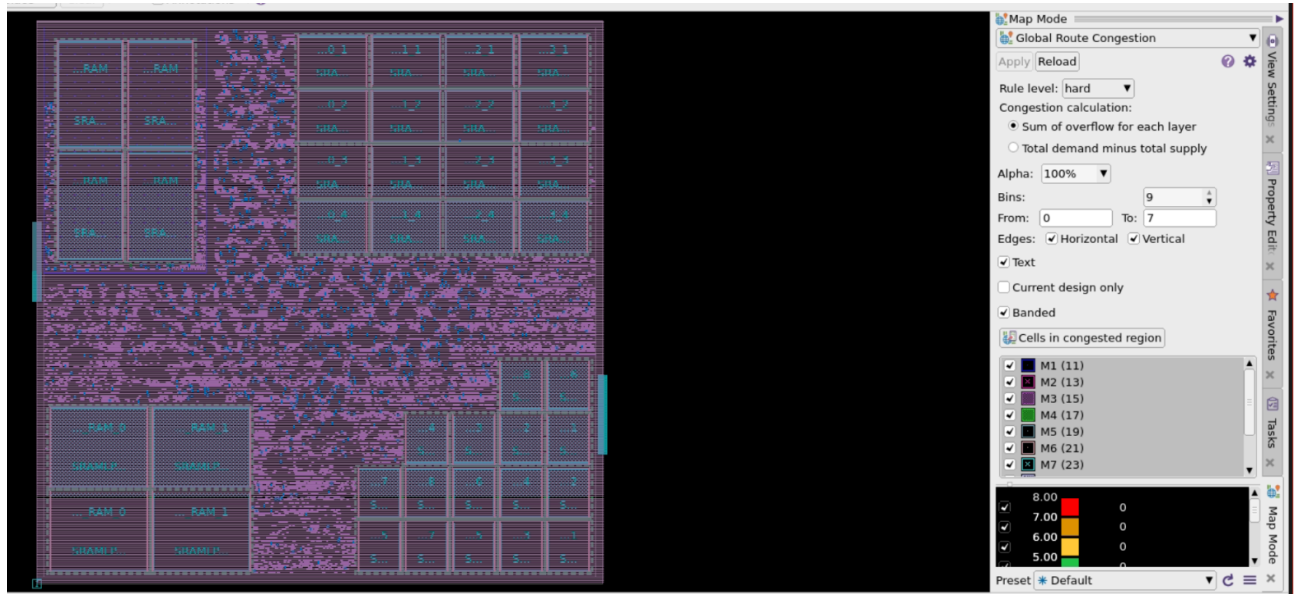


Figure 15: Global Route Congestion

- **Scan Chain:**

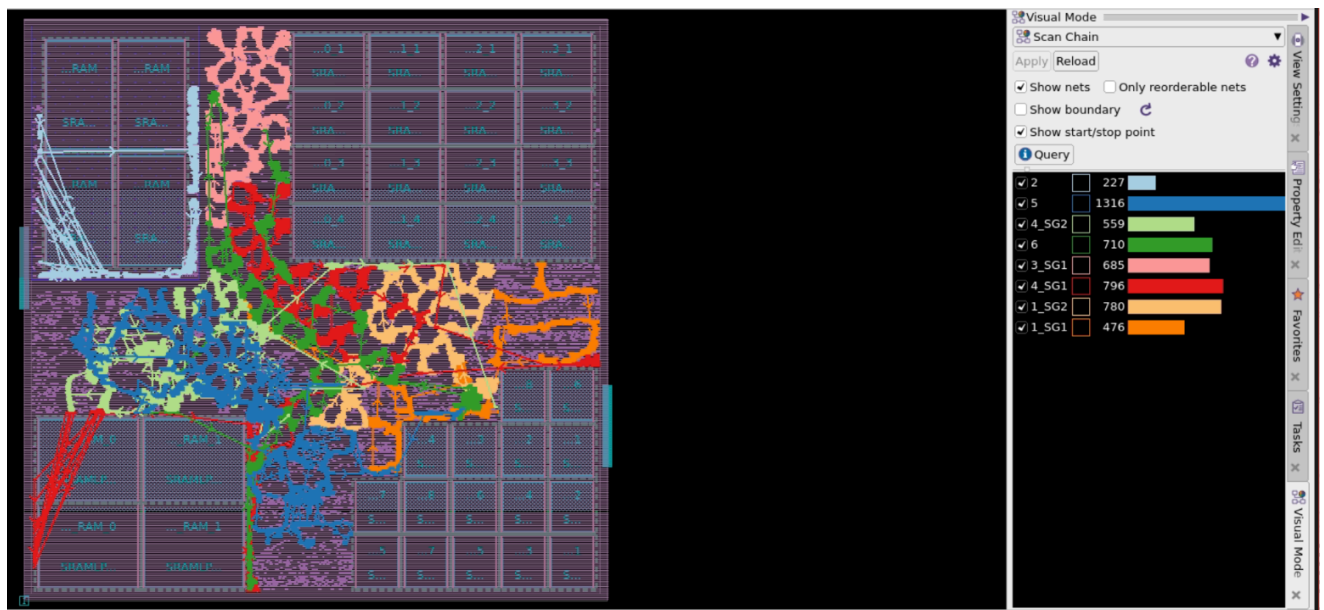


Figure 16: Scan Chain

3.3.3 Sanity check (Placement)

- **Report Constraints:** Strict rules were applied to **max_transition** and **max_capacitance** to stabilize timing paths.

```
fc_shell> report_constraint
*****
Report : constraint
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Tue Feb 24 04:45:18 2026
*****

Constraint                                     Cost
-----
min_delay/hold                                0.69 (VIOLATED)
max_delay/setup                               0.35 (VIOLATED)
max_transition                                46.94 (VIOLATED)
max_capacitance                              251.38 (VIOLATED)
min_capacitance                              0.00 (MET)
```

Figure 17: Report constraints

- **Global Core Utilization: 0.7289**

```
report_utilization -config util_config
*****
Report : report_utilization
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Tue Feb 24 04:43:36 2026
*****
Utilization Ratio:                                0.7289
Utilization options:
- Area calculation based on:      core_area of block ORCA_TOP
- Categories of objects excluded: hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                        586645.6781
Total Capacity Area:               254043.9686
Total Area of cells:               185185.3295
Area of excluded objects:
- hard_macros      :      264884.2695
- macro_keepouts  :      67717.4400
- soft_macros     :       0.0000
- io_cells        :       0.0000
- hard_blockages  :       0.0000
Total Area of excluded objects:    332601.7095
Ratio of excluded objects:         0.5670

0.7289
```

Figure 18: Global Core Utilization Report

- **RISC Core Utilization: 0.5404**

```
report_utilization -config util_config -of_objects [get_voltage_areas PD_RISC_CORE]
*****
Report : report_utilization
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Tue Feb 24 04:44:04 2026
*****
Utilization Ratio:                0.5404
Utilization options:
- Area calculation based on:      Voltage area PD_RISC_CORE
- Categories of objects excluded: hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                       77050.1830
Total Capacity Area:              16119.3000
Total Area of cells:              8711.0397
Area of excluded objects:
- hard_macros      :              51184.6830
- macro_keepouts  :              9746.2000
- soft_macros     :               0.0000
- io_cells        :               0.0000
- hard_blockages  :               0.0000
Total Area of excluded objects:    60930.8830
Ratio of excluded objects:         0.7908

0.5404
```

Figure 19: RISC Core Utilization Report

- **Report timing:** Setup violation WNS max at -1.75; Hold violation WNS max at -0.18

Timing			
Context	WNS	TNS	NVE

func_ss0p75v125c_cmax			
(Setup)	-1.75	-672.21	1213
test_ss0p75v125c_cmax			
(Setup)	-0.35	-1.03	3
Design	-1.75	-672.21	1213
func_ss0p75v125c_cmax			
(Hold)	-0.18	-3.38	124
test_ss0p75v125c_cmax			
(Hold)	-0.20	-32.65	424
Design	-0.20	-32.71	425

Figure 20: Timing Report

3.4 Clock Tree Synthesis (CTS)

3.4.1 Clock Tree Synthesis Implementation

Clock Tree Synthesis was executed under dedicated CTS and ClockOpt scenarios with multi-corner optimization enabled. The clock network was built using the `clock_opt` flow, routing primarily on **M5–M8** with LVT buffers and clock gating cells only. CRPR removal and local skew optimization were applied, targeting **100 ps skew** and **0.15 ns max transition**.

A custom **2w2s NDR** increased metal width and spacing to enhance signal integrity and clock stability. Post-CTS optimization extended routing to **M1–M9**, enabling high-effort hold fixing and congestion-aware refinement. Implementation controls included tie cell constraints (fanout = 8), CTS cell filtering, defined instance prefixes (**CTS_**, **CLKOPT_**), and automatic PG reconnection.



Figure 21: Clock path routed

- Clock Tree Structure of ate_clk

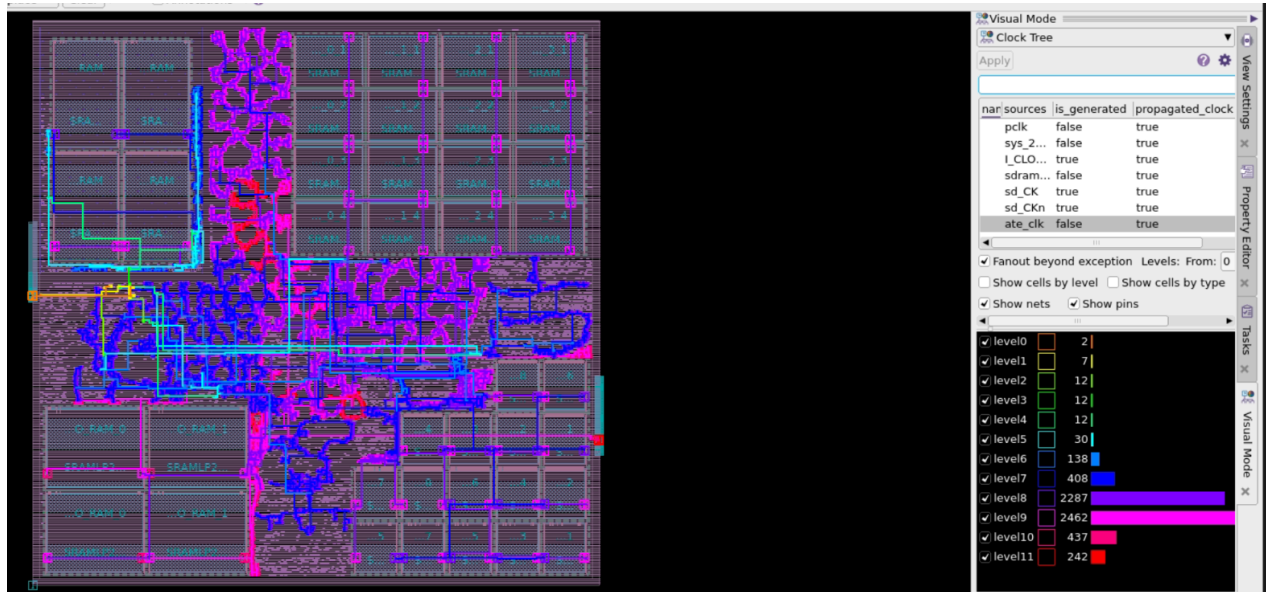


Figure 22: ate_clk Clock Tree

- Leveled Clock Tree of ate_clk

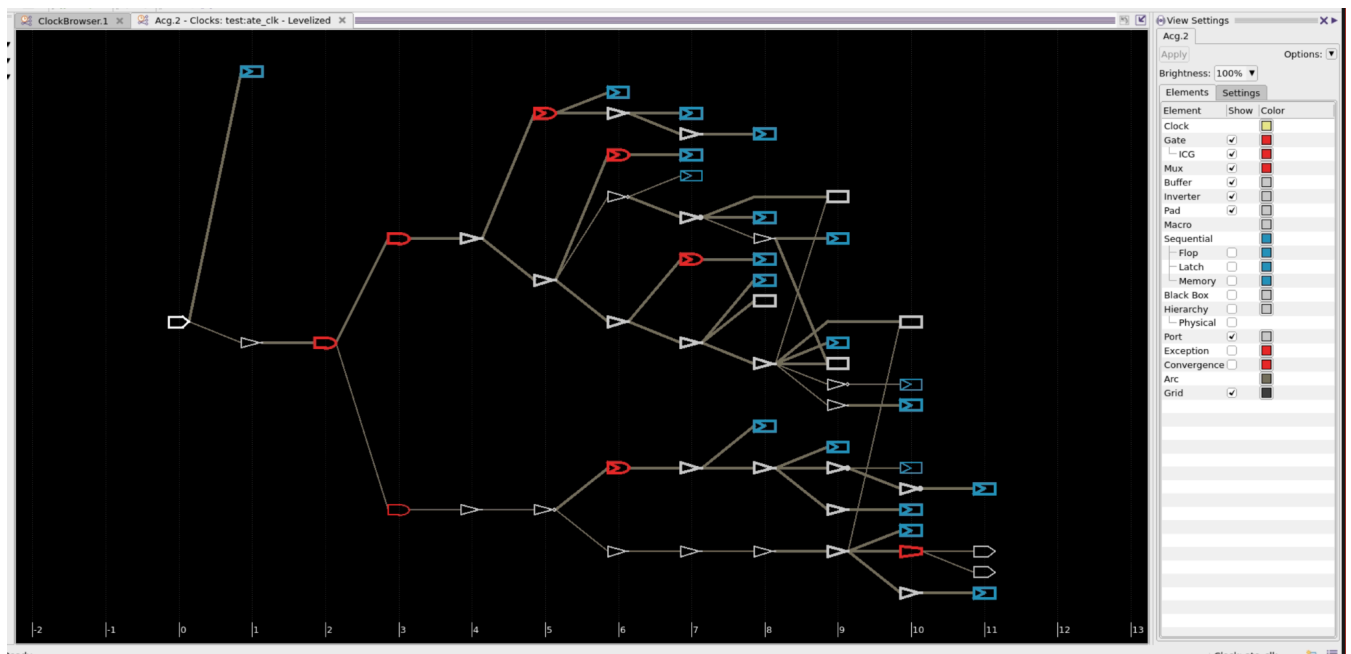


Figure 23: ate_clk Leveled Clock Tree

3.4.2 Sanity check (Clock Tree Synthesis)

- **Report Constrains:**

```
fc_shell> report_constraint
*****
Report : constraint
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Wed Feb 25 07:10:04 2026
*****
```

Constraint	Cost
min_delay/hold	0.46 (VIOLATED)
max_delay/setup	0.00 (MET)
max_transition	111.10 (VIOLATED)
max_capacitance	1061.41 (VIOLATED)
min_capacitance	0.00 (MET)

Figure 24: Constrains

- **Report timing:** Both Setup/Hold timing violation is increased after CTS.

Timing			
Context	WNS	TNS	NVE
func_ff0p95vm40c_cmin			
(Setup)	-2.37	-1142.46	1373
func_ss0p75v125c_cmax			
(Setup)	-1.44	-287.17	368
test_ff0p95v125c_cmin			
(Setup)	0.03	0.00	0
test_ss0p75v125c_cmax			
(Setup)	0.03	0.00	0
Design (Setup)	-2.37	-1143.12	1379
func_ff0p95vm40c_cmin			
(Hold)	-0.21	-2.12	28
func_ss0p75v125c_cmax			
(Hold)	-0.03	-0.32	21
test_ff0p95v125c_cmin			
(Hold)	-0.43	-24.38	150
test_ss0p75v125c_cmax			
(Hold)	-0.43	-24.38	150
Design (Hold)	-0.43	-24.38	152

Figure 25: Timing Report

- **Report Congestions: 0.15% overflow**

```

fc_shell> report_congestion
*****
Report : congestion
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Wed Feb 25 08:44:46 2026
*****

Layer   |   overflow   |           # GRCs has
Name    |  total  | max  | overflow (%) | max overflow
-----|-----|-----|-----|-----
Both Dirs |    658 |    2 |    655 ( 0.15%) |    3
H routing |    657 |    2 |    654 ( 0.29%) |    3
V routing |     1  |    1 |     1 ( 0.00%) |    1

```

Figure 26: Congestions Report

3.5 Routing & Post-Route Optimization

After Clock Tree Synthesis, the design proceeded to detailed routing using the full **9-metal layer stack** under DRC and antenna constraints. Dedicated Route and RouteOpt scenarios were configured with multi-corner timing, propagated clock analysis, and simultaneous setup/hold evaluation enabled.

A hierarchical routing strategy was applied, reserving **M1–M3** for local interconnections and **M4–M9** for global and clock routing. Timing-driven and crosstalk-driven routing were activated to address signal integrity concerns under 75% core utilization. Clock nets were routed on **M5–M8** using a 2w2s Non-Default Rule (NDR) to enhance stability.

Post-route parasitic RC extraction initially degraded setup timing. High-effort RouteOpt optimization achieved approximately **0.9 ns setup recovery**, ensuring compliance with the **16.7 ns clock period** target. Although significant timing improvement was achieved, some DRC and LVS violations remained at this stage, indicating that further refinement is required for full signoff closure.

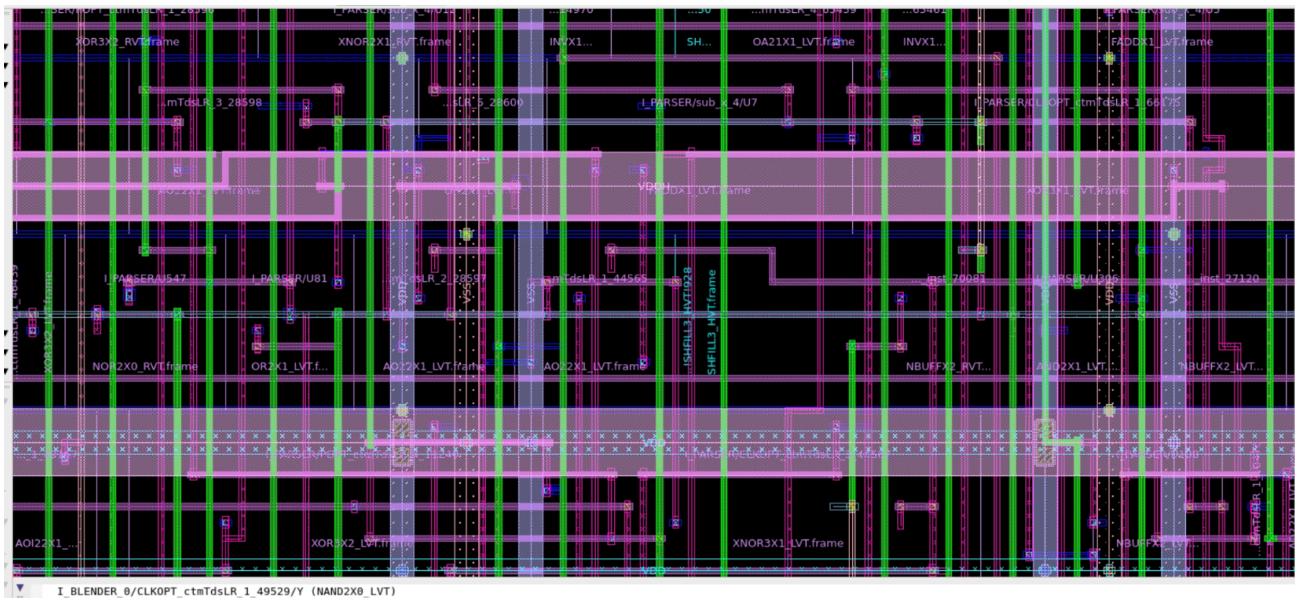


Figure 27: Routed net

4. Final Quality of Results (QoR) Analysis

The final implementation was verified against industry-standard sign-off criteria:

- **Congestion Analysis:** The final `report_congestion` status is "Safe", despite the 75% utilization rate, proving the effectiveness of our layer assignment strategy.

```
fc_shell> report_congestion
*****
Report : congestion
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Wed Feb 25 08:44:46 2026
*****

Layer  |  overflow  |          # GRCs has
Name   |  total   | max   | overflow (%) | max overflow
-----|-----|-----|-----|-----
Both Dirs |    658 |    2 |    655 ( 0.15%) |    3
H routing |    657 |    2 |    654 ( 0.29%) |    3
V routing |     1 |    1 |     1 ( 0.00%) |    1
```

Figure 28: Congestion Report

- **Timing Report:**

```
fc_shell> report_global_timing
Information: Timer using 'CRPR'. (TIM-050)
*****
Report : global timing
        -format { narrow }
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Wed Feb 25 19:26:33 2026
*****

Setup violations
-----
              Total  reg->reg  in->reg  reg->out  in->out
-----
WNS          -1.46    -1.46     0.00     0.00     0.00
TNS        -290.18   -290.18     0.00     0.00     0.00
NUM             358     358         0         0         0
-----

Hold violations
-----
              Total  reg->reg  in->reg  reg->out  in->out
-----
WNS          -0.03    -0.02    -0.03     0.00     0.00
TNS          -0.22    -0.17    -0.04     0.00     0.00
NUM             18     16         2         0         0
-----
```

Figure 29: Global timing Report

- **Physical Integrity:** Parasitic estimation via `report_parasitics` confirmed that the RC values align with the expected limits for the 32nm process.

```
*****
Report : parasitics
        -late
        -rise
Module : ORCA_TOP
Mode   : test
Corner : ss0p75v125c_cmax
Scenario: test_ss0p75v125c_cmax
Version: V-2023.12-SP4
Date   : Wed Feb 25 20:21:29 2026
*****

Corner : ss0p75v125c_cmax

cap units: 1.00fF  res units: 1.00M0hm  time units: 1.00ns

Current delay calculation style: auto
Current min Elmore tau: 0.00s
Current min Arnoldi tau: 2.00ps

Total nets: 58144  RC networks: 58144 (100.00%)

Grand total early parasitic network R: 2.581700  C: 119374.891088
Grand total early lumped parasitic C: 119374.891064
Grand total late parasitic network R: 2.581700  C: 119374.891088
Grand total late lumped parasitic C: 119374.891064

Average pins per network: 3.73
Average caps per network: 19.05
Average xCaps per network: 16.08
Average resistors per network: 18.05
Cap/pin ratio: 5.11
```

Figure 30: Parasitic report

- **Checking Legality:**

```
===== 2D Rule Check Legality Report =====
                        Violating Cells
Legality Rules          Moveable AppFixed UserFixed   Total
-----
overlap                  0           0           0         0
spacing_rule             0           0           0         0
cell_on_site             0           0           0         0
legal_orient             0           0           0         0
va_bound                 0           0           0         0
pg_drc                   0           0           0         0
-----
Total Violations         0           0           0         0
=====
Information: Dynamic Rule Ordering (DRO): true
```

Figure 31: Legality check after Routeopt

- Check multi-voltage design:

```

*****
Report : check_mv_design
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Wed Feb 25 22:35:36 2026
*****

----- Power domain rule -----
No errors or warnings.

----- Supply set rule -----
No errors or warnings.

----- Supply net rule -----
No errors or warnings.

----- Supply port rule -----
No errors or warnings.

----- Isolation strategy rule -----
No errors or warnings.

----- Level shifter strategy rule -----
No errors or warnings.

----- Retention strategy rule -----
No errors or warnings.

----- Power switch strategy rule -----
No errors or warnings.

----- Repeater rule -----
No errors or warnings.

----- Analog net rule -----
No errors or warnings.

----- Physical block pin rule -----
No errors or warnings.

----- PG pin rule -----
No errors or warnings.

----- Signal pin rule -----
No errors or warnings.

----- Summary -----
Information: Total 0 error(s) and 0 warning(s) from check_mv_design. \(MV-082\)

```

Figure 32: Mv design check after Routeopt

- **Report Constraints:** 0.62 min_delay/hold violated, 104.91 max_transition violated, 1089.75 max_capacitance violated

```
report_constraints
*****
Report : constraint
Design : ORCA_TOP
Version: V-2023.12-SP4
Date   : Wed Feb 25 22:42:46 2026
*****
```

Constraint	Cost
min_delay/hold	0.62 (VIOLATED)
max_delay/setup	0.00 (MET)
max_transition	104.91 (VIOLATED)
max_capacitance	1089.75 (VIOLATED)
min_capacitance	0.00 (MET)

Figure 33: Constrains report after Routeopt

- **Checking routes:** 8306 DRCs errors left.

```
Verify Summary:

Total number of nets = 57332, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                0 ports without pins of 0 cells connected to 0 nets
                                0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 8306
Total number of antenna violations = no antenna rules defined
Information: Routes in non-preferred voltage areas = 256 (ZRT-559)
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked
```

Figure 34: Routes check after Routeopt

- **Checking LVS errors:** 20 short violations and 3 open nets left.

```
Total number of input nets is 57516.
Total number of short violations is 20.
Total number of open nets is 3.
Open nets are VDD VDDH VSS
Total number of floating route violations is 20.
```

```
Elapsed = 0:00:37, CPU = 0:00:36
```

Figure 35: Lvs check after Routeopt

- **Checking Power/Ground missing vias:** 25 missing vias left.

```
fc_shell> check_pg_missing_vias
Check net VDD vias...
Number of missing vias on VIA1 layer: 1
Total number of missing vias: 1
Checking net VDD vias took 1 seconds.
Check net VDDH vias...
Number of missing vias on VIA5 layer: 24
Total number of missing vias: 24
Checking net VDDH vias took 1 seconds.
Check net VSS vias...
Number of missing vias: 0
Checking net VSS vias took 2 seconds.
Overall runtime: 5 seconds.
```

Figure 36: Missing vias check after Routeopt

5. Conclusion

The physical implementation of the ORCA_TOP RISC-V core was successfully carried out, progressing from a gate-level netlist to a routed physical layout using the SAED 32nm technology library and Synopsys Fusion Compiler. The project demonstrates a complete ASIC backend flow, including floorplanning, multi-voltage planning, placement optimization, clock tree synthesis, and detailed routing.

An R-shaped floorplan enabled 75% core utilization while preserving placement legality and routing feasibility. A hierarchical power mesh implemented across upper metal layers ensured stable power delivery within dual-voltage domains. During Clock Tree Synthesis, the *ate_clk* network was balanced with a 100 ps skew target, and post-route SI-aware optimization recovered approximately 0.9 ns of setup slack, satisfying the 16.7 ns clock period requirement.

Post-route analysis indicates acceptable congestion distribution and density control under relatively high utilization. Although minor DRC and LVS violations remain and require further refinement for full signoff closure, the design achieves stable timing behavior and demonstrates practical convergence of PPA objectives within a realistic physical design environment.

6. References

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